BASIC ELECTRICAL ENGINEERING: COMPREHENSIVE MODULE

1. DC Circuit Analysis: Beyond Ohm's Law

Network Theorems

Thevenin’s Theorem: Any linear bilateral network can be replaced by an equivalent circuit containing a voltage source (V_{th}) in series with a resistor (R_{th}). This is essential for simplifying complex circuits.

Norton’s Theorem: Similar to Thevenin, but uses a current source (I_n) in parallel with a resistor (R_n).

Superposition Theorem: In a circuit with multiple sources, the current in any branch is the algebraic sum of currents produced by each source acting independently.

2. Single Phase AC Circuits

Complex Impedance &

AC signals are represented as rotating phasors. Impedance (Z) is defined as $R + jX$, where X reactance

$$Z = \text{sqrt}(R^2 + \angle \quad ^{-1}(X/R$$

Power Crucial for efficiency, defined as the cosine of the phase angle between and current. Unity power factor ($\cos\varphi=1$) is

3. Transformers: Principles of Induction

The Ideal Transformer

Transformers work on mutual induction. The transformation ratio (k) determines the voltage step-up or step-down characteristic.

$$k = N2/N1 = V2/V1 = I1/I2$$

Losses: Iron losses (hysteresis and eddy current) and Copper losses (I^2R loss) constitute total power loss.

4. Advanced Problem Bank

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Detailed Problem: Thevenin Equivalent
Find the Thevenin equivalent circuit across terminals A and B for a circuit with a 10V source, 2Ω resistor, and 4Ω load.

> **Detailed Solution:**
> 1. Remove load (4Ω). Find Open Circuit Voltage (Voc).
> 2. Vth = 10V * (R2 / (R1+R2)) -- if using voltage divider.
> 3. For a simple series circuit, Vth = 10V.
> 4. Find Rth: Short circuit the voltage source. Rth = 2Ω.
> 5. Final circuit: 10V source in series with 2Ω resistor connected to load.

5. Electrical Machines: DC & AC

Electromechanical Energy Conversion

DC Generator: Works on Faraday's Law; converts mechanical energy to electrical energy using commutators to maintain DC output.

Induction Motor: Works on the principle of a rotating magnetic field inducing a current in the rotor, resulting in torque.